# Numerical Differentiation

**Chapter 02**

**Lesson 1**

# 1. Introduction to Numerical Differentiation

- In calculus, the derivative of a function $f(x)$is defined as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

- This formula requires the function $f(x)$to be known.

- However, in many practical applications, the function may not be available explicitly. Instead, we may have a set of **tabulated or experimental data**.

- For example:

| $x$ | $x_0$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $f(x)$ | $f_0$ | $f_1$ | $f_2$ | $f_3$ | $f_4$ |

- We may need to estimate

$$f'(x), f''(x), f'''(x), \ldots$$

from these data values.

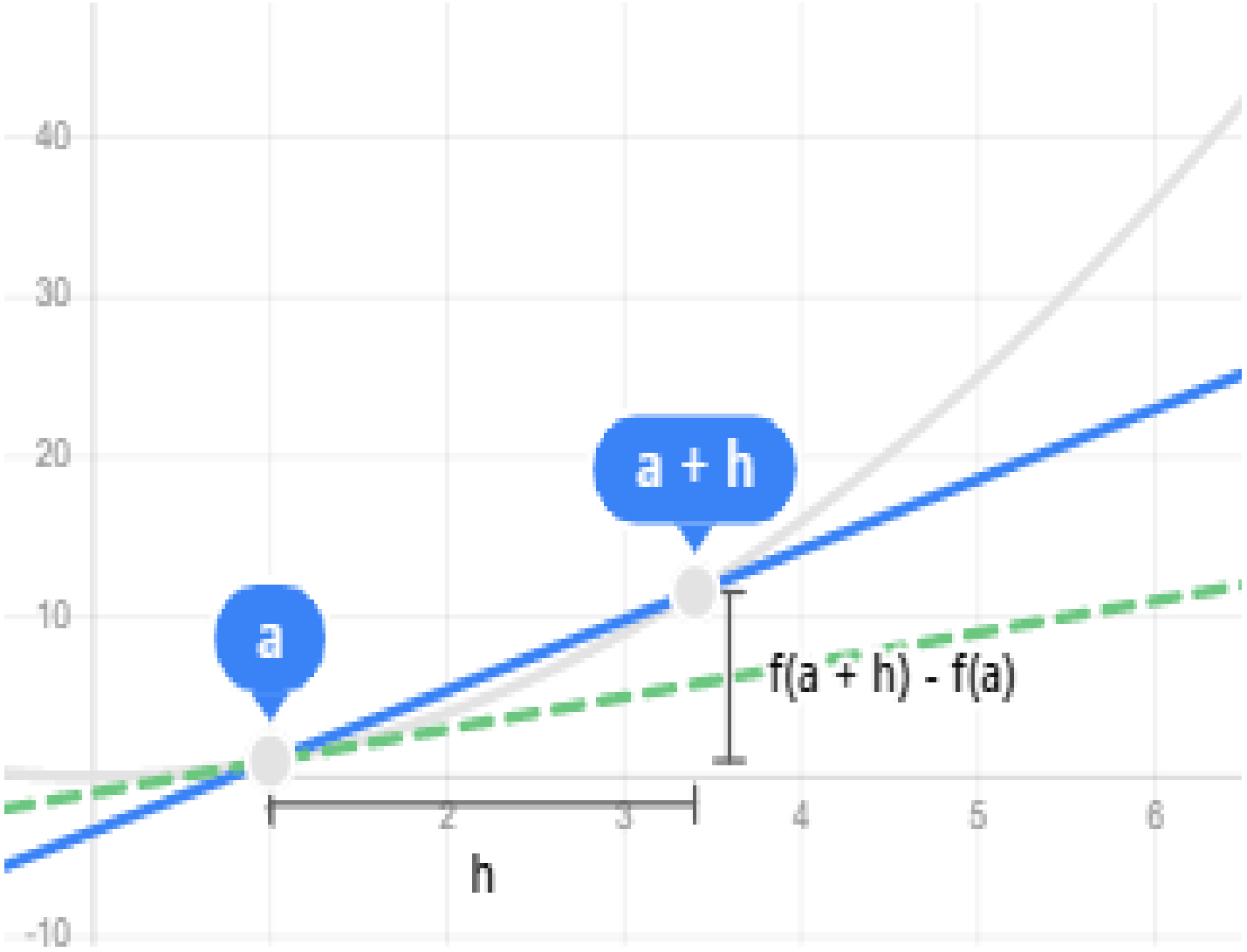
- This process is called **numerical differentiation**.

# Why is numerical differentiation needed?

- It is useful when:

1. The function is not available explicitly.
2. Only experimental or observed data are available.
3. The function is complicated to differentiate analytically.
4. Derivatives need to be estimated from a data table.
5. Measurements are obtained at discrete points.

# Derivative as a Slope



The first derivative represents the slope of a curve.

For two nearby points,

$$(x, f(x))$$

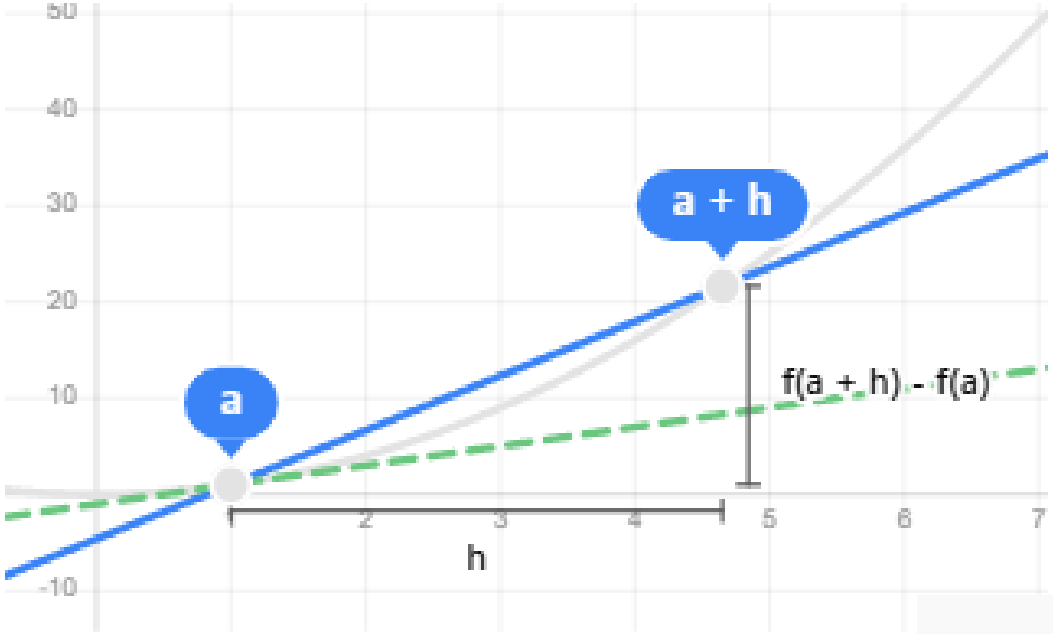
and

$$x + h, f(x + h)$$

the slope of the secant line is

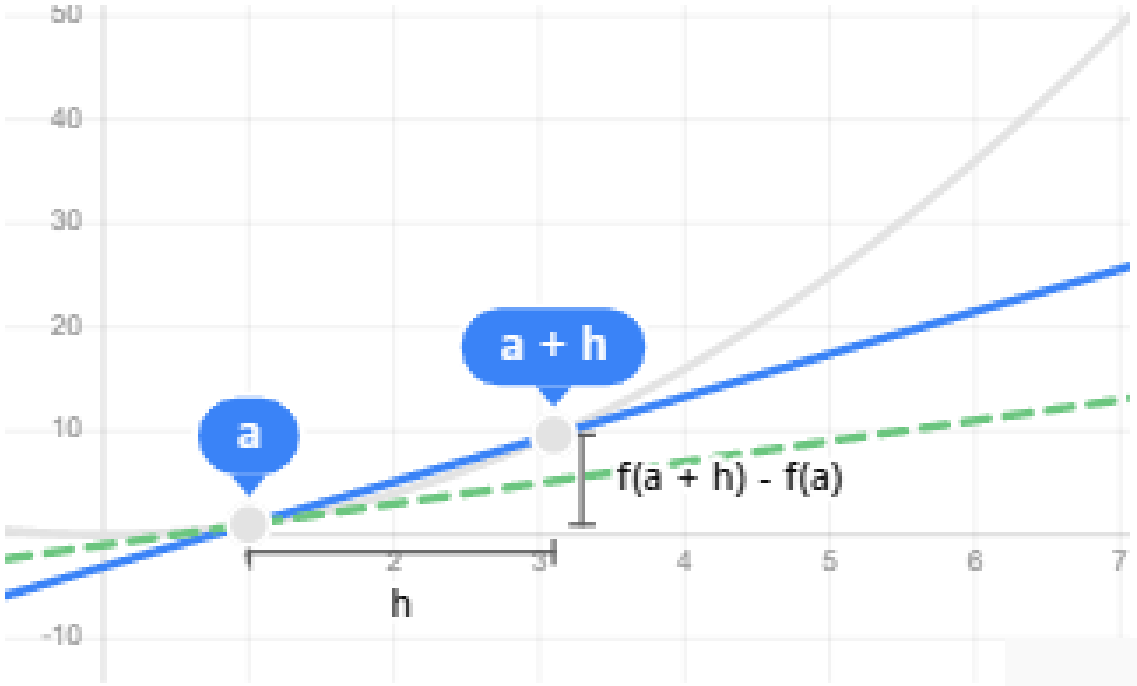
$$\frac{f(x+h) - f(x)}{h}.$$

As $h$ becomes smaller, the secant slope approaches the tangent slope.



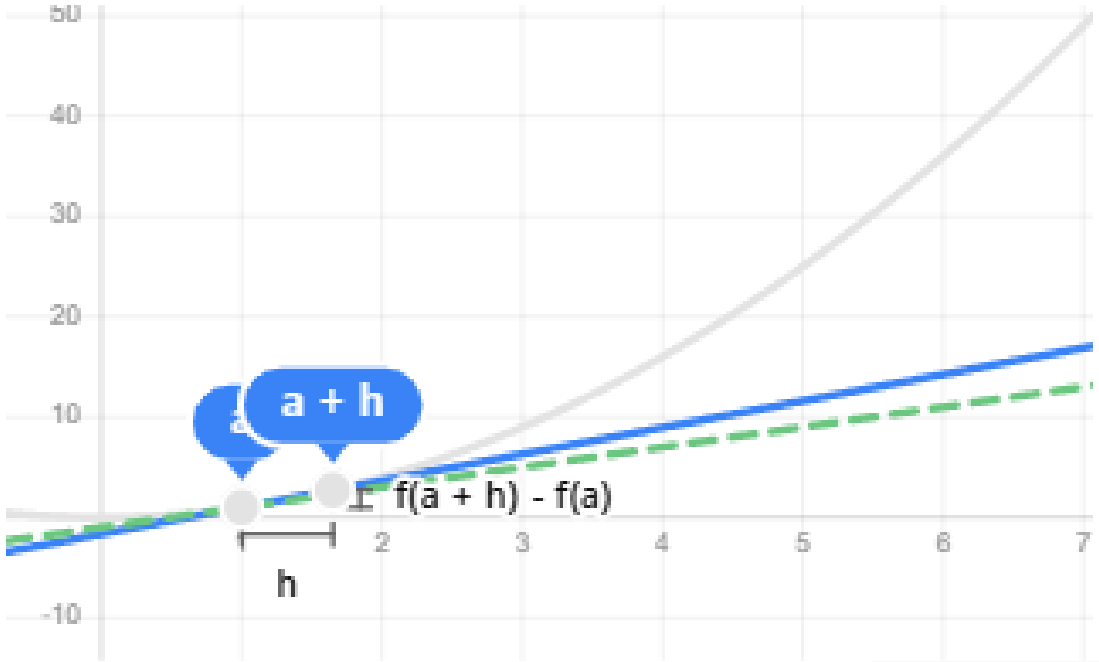

$$f'(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$

$$\frac{f(1+3.65) - f(1)}{3.65} = 5.65 \quad \text{error} = 3.65$$



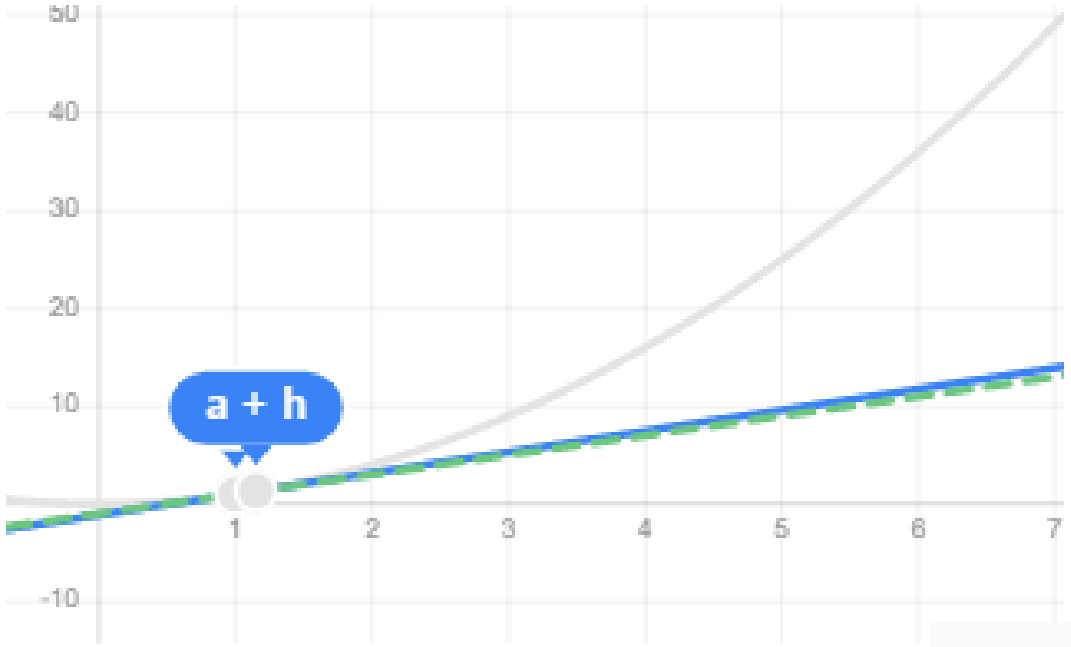

$$f'(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$

$$\frac{f(1+2.1) - f(1)}{2.1} = 4.1 \quad \text{error} = 2.1$$




$$f'(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$

$$\frac{f(1+0.65) - f(1)}{0.65} = 2.65 \quad \text{error} = 0.65$$




$$f'(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$

$$\frac{f(1+0.15) - f(1)}{0.15} = 2.15 \quad \text{error} = 0.15$$

# 2. Finite Difference Concept

- A **finite difference** approximates the change in a function between two or more discrete points.

Suppose the values of $x$are equally spaced:

$$x_0, x_1, x_2, \ldots$$

Where,

$$x_{i+1} - x_i = h.$$

Here, $h$ is called the **step size** or **grid spacing**.

Thus,

$$x_1 = x_0 + h$$
$$x_2 = x_0 + 2h$$
$$x_3 = x_0 + 3h.$$

Let

$$f_i = f(x_i).$$

- Then the data can be represented as

| $x$ | $x_0$ | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|---|
| $f(x)$ | $f_0$ | $f_1$ | $f_2$ | $f_3$ | $f_4$ |

The main types of finite differences are:

1. **Forward difference**
2. **Backward difference**

# 3. Forward Difference Operator

## Learning Objectives

**At the end of this lecture, students should be able to:**

- Explain the concept of numerical differentiation.
- Define finite differences and the forward difference operator.
- Construct a forward difference table.
- Derive forward difference formulas using Taylor series.
- Apply first-, second-, and higher-order forward difference formulas.
- Calculate truncation errors.

- The forward difference operator is denoted by

$$\boxed{\Delta}$$

- The first forward difference of $f(x)$ is defined as

$$\boxed{\Delta f(x) = f(x+h) - f(x)}$$

- For tabulated data,

$$\boxed{\Delta f_i = f_{i+1} - f_i}$$

**Example**

Consider:

| $x$ | $f(x)$ |
|---|---|
| 0 | 2 |
| 1 | 5 |
| 2 | 10 |
| 3 | 17 |

The first differences are:

$$\Delta f_0 = f_1 - f_0$$
$$= 5 - 2 = 3$$

Similarly,

$$\Delta f_1 = 10 - 5 = 5$$

and

$$\Delta f_2 = 17 - 10 = 7.$$

- Therefore,

$$\boxed{\Delta f_0 = 3, \Delta f_1 = 5, \Delta f_2 = 7}$$

# 4. First-Order Forward Difference Formula

The simplest forward difference formula for the first derivative is

$$\boxed{f'(x_i) \approx \frac{f_{i+1} - f_i}{h}}$$

or equivalently,

$$\boxed{f'(x) \approx \frac{f(x+h) - f(x)}{h}}$$

This is called the **first-order forward difference formula**.

**Why is it called first-order?**

Because its truncation error is

$$\boxed{O(h)}$$

which means that the error is approximately proportional to $h$.

# Example 1: First-Order Forward Difference

**Given**

$$\boldsymbol{f(x) = x^2 + 2x}$$

**estimate $\boldsymbol{f'(1)}$ using**

$$h = 0.1.$$

Step 1: Formula

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Therefore,

$$f'(1) \approx \frac{f(1.1) - f(1)}{0.1}.$$

**Step 2: Calculate $f(1)$**

$$f(1) = 1^2 + 2(1) = 3.$$

**Step 3: Calculate $f(1.1)$**

$$f(1.1) = (1.1)^2 + 2(1.1)$$
$$= 1.21 + 2.2$$
$$= 3.41.$$

**Step 4: Substitute**

$$f'(1) \approx \frac{3.41 - 3}{0.1}$$
$$= \frac{0.41}{0.1}$$

$$\boxed{f'(1) \approx 4.1}$$

**Exact answer**

$$f'(x) = 2x + 2$$

Therefore,

$$f'(1) = 4.$$

Hence,

$$\boxed{\text{Absolute error} = |\ 4.1 - 4\ | = 0.1}$$

# 5. Second-Order Forward Difference Formula

The term **second-order forward difference formula** can refer to the formula for the **second derivative**.

We want to estimate

$$f''(x).$$

The formula is

$$\boxed{f''(x_i) \approx \frac{f_{i+2} - 2f_{i+1} + f_i}{h^2}}$$

or

$$\boxed{f''(x) \approx \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}}$$

Notice that

$$\Delta^2 f_i = f_{i+2} - 2f_{i+1} + f_i.$$

How it happen?

The **second forward difference** is defined as

$$\boxed{\Delta^2 f_i = \Delta(\Delta f_i)}$$

We already know:

$$\Delta f_i = f_{i+1} - f_i$$

Now, to calculate the forward difference of this quantity, we move one step forward:

$$\Delta f_{i+1} = f_{i+2} - f_{i+1}$$

Therefore,

$$\Delta^2 f_i = \Delta f_{i+1} - \Delta f_i$$

Substitute the two expressions:

$$\Delta^2 f_i = (f_{i+2} - f_{i+1}) - (f_{i+1} - f_i)$$

Expand the brackets:

$$\Delta^2 f_i = f_{i+2} - f_{i+1} - f_{i+1} + f_i$$

Combine the middle terms:

$$\boxed{\Delta^2 f_i = f_{i+2} - 2f_{i+1} + f_i}$$

Therefore,

$$\boxed{f''(x_i) \approx \frac{\Delta^2 f_i}{h^2}}$$

The truncation error is

$$\boxed{O(h)}$$

for this basic formula.

**Example 2: Second Derivative**

Let

$$f(x) = x^3 + 2x^2 + x.$$

Estimate

$$f''(1)$$

using

$$h = 0.1.$$

**Step 1: Formula**

$$f''(1) \approx \frac{f(1.2) - 2f(1.1) + f(1)}{(0.1)^2}.$$

**Step 2: Calculate $f(1)$**

$$f(1) = 1^3 + 2(1)^2 + 1 = 4.$$

**Step 3: Calculate $f(1.1)$**

$$\begin{aligned} f(1.1) &= (1.1)^3 + 2(1.1)^2 + 1.1 \\ &= 1.331 + 2.42 + 1.1 \\ &= 4.851. \end{aligned}$$

**Step 4: Calculate $f(1.2)$**

$$\begin{aligned} f(1.2) &= (1.2)^3 + 2(1.2)^2 + 1.2 \\ &= 1.728 + 2.88 + 1.2 \\ &= 5.808. \end{aligned}$$

**Step 5: Substitute**

$$f''(1) \approx \frac{5.808 - 2(4.851) + 4}{0.01}$$

$$= \frac{5.808 - 9.702 + 4}{0.01}$$

$$= \frac{0.106}{0.01}$$

$$\boxed{f''(1) \approx 10.6}$$

**Exact result**

$$f''(x) = 6x + 4$$

so

$$f''(1) = 10.$$

- Thus,

$$\boxed{\text{Absolute error} = 0.6}$$

# 6. Higher-Order Forward Difference Formulas

Higher-order forward differences are obtained by repeatedly applying the forward difference operator.

**First Difference**

$$\boxed{\Delta f_i = f_{i+1} - f_i}$$

**Second Difference**

$$\Delta^2 f_i = \Delta(\Delta f_i)$$

$$\boxed{\Delta^2 f_i = f_{i+2} - 2f_{i+1} + f_i}$$

**Third Difference**

$$\boxed{\Delta^3 f_i = f_{i+3} - 3f_{i+2} + 3f_{i+1} - f_i}$$

**Fourth Difference**

$$\boxed{\Delta^4 f_i = f_{i+4} - 4f_{i+3} + 6f_{i+2} - 4f_{i+1} + f_i}$$

# 7. General Forward Difference Formula

The general $n$-th forward difference is

$$\Delta^n f_i = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f_{i+k}$$

Where,

- $n$= order of the difference,
- $k$= index,
- $\binom{n}{k}$= binomial coefficient.

For example, when $n = 3$:

$$\Delta^3 f_i = \sum_{k=0}^{3} (-1)^{3-k} \binom{3}{k} f_{i+k}.$$

This gives

$$\Delta^3 f_i = -f_i + 3f_{i+1} - 3f_{i+2} + f_{i+3}.$$

Therefore,

$$\boxed{\Delta^3 f_i = f_{i+3} - 3f_{i+2} + 3f_{i+1} - f_i}$$

# 8. Derivation Using Taylor Series

Taylor series is used to derive finite difference formulas.

Suppose $f(x)$ is sufficiently differentiable.

The Taylor expansion around $x$ is

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots$$

or

$$\boxed{f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots}$$

**Derivation of First-Order Formula**

Starting with

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots$$

Subtract $f(x)$:

$$f(x+h) - f(x) = hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots$$

Divide by $h$:

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + \frac{h^2}{6}f'''(x) + \cdots$$

Therefore,

$$\boxed{f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{h}{2}f''(x) + O(h^2)}$$

Ignoring the higher-order terms gives

$$\boxed{f'(x) \approx \frac{f(x+h) - f(x)}{h}}$$

# 9. Derivation of Three-Point Forward Formula

A more accurate formula for the first derivative can be obtained using three points:

$$x, x+h, x+2h.$$

Taylor expansion gives

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots$$

And (Now replace $h$by $2h$)

$$f(x+2h) = f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4}{3}h^3f'''(x) + \cdots$$

Multiply the first equation by 4:

$$4f(x+h) = 4f(x) + 4hf'(x) + 2h^2f''(x) + \frac{2}{3}h^3f'''(x) + \cdots$$

Now calculate

$$4f(x+h) - f(x+2h).$$

After simplification,

$$4f(x+h) - f(x+2h) = 3f(x) + 2hf'(x) + O(h^3).$$

Therefore,

$$-3f(x) + 4f(x+h) - f(x+2h) = 2hf'(x) + O(h^3).$$

Dividing by $2h$:

$$\boxed{f'(x) \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}}$$

The truncation error is

$$\boxed{O(h^2)}$$

which is more accurate than the basic forward difference formula.

**Example 3: Three-Point Forward Difference**

**Given**

$$f(x) = x^2 + 2x$$

**estimate**

$$f'(1)$$

**using**

$$h = 0.1.$$

**Step 1**

Use

$$f'(x) \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}.$$

**Step 2**

We have

$$f(1) = 3,$$
$$f(1.1) = 3.41,$$

and

$$f(1.2) = 3.84.$$

**Step 3**

Substitute:

$$f'(1) \approx \frac{-3(3) + 4(3.41) - 3.84}{2(0.1)}$$
$$= \frac{-9+13.64-3.84}{0.2} = \frac{0.8}{0.2}$$
$$\boxed{f'(1) = 4}$$

The exact derivative is

$$f'(x) = 2x + 2$$

so

$$f'(1) = 4.$$

Therefore,

$$\boxed{\text{Error} = 0}$$

for this particular example

**Example Using a Forward Difference Table**

Given:

| $x$ | $f(x)$ |
|---|---|
| $0$ | $1$ |
| $1$ | $4$ |
| $2$ | $9$ |
| $3$ | $16$ |
| $4$ | $25$ |

Construct the forward difference table.

**First differences**

$$\Delta f_0 = 4 - 1 = 3$$
$$\Delta f_1 = 9 - 4 = 5$$
$$\Delta f_2 = 16 - 9 = 7$$
$$\Delta f_3 = 25 - 16 = 9$$

**Second differences**

$$\Delta^2 f_0 = 5 - 3 = 2$$
$$\Delta^2 f_1 = 7 - 5 = 2$$
$$\Delta^2 f_2 = 9 - 7 = 2$$

**Third differences**

$$\Delta^3 f_0 = 2 - 2 = 0$$
$$\Delta^3 f_1 = 2 - 2 = 0.$$

Therefore:

| $x$ | $f(x)$ | $\Delta f$ | $\Delta^2 f$ | $\Delta^3 f$ |
|---|---|---|---|---|
| $0$ | $1$ | $3$ | $2$ | $0$ |
| $1$ | $4$ | $5$ | $2$ | $0$ |
| $2$ | $9$ | $7$ | $2$ | |
| $3$ | $16$ | $9$ | | |
| $4$ | $25$ | | | |

Since the third differences are zero, the data are generated by a quadratic polynomial.

# 12. Error / Truncation Error

Numerical formulas are approximations. Therefore, it is important to understand their errors.

**First-Order Forward Difference**

From Taylor expansion:

$$\frac{f(x+h)-f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + O(h^2).$$

Therefore,

$$\boxed{\text{Truncation error} = O(h)}$$

# Explanation

**1. Start with Taylor expansion**

We have

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \cdots$$

Move $f(x)$to the left:

$$f(x+h) - f(x) = hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots$$

Now divide everything by $h$:

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + \frac{h^2}{6}f'''(x) + \cdots$$

Therefore,

$$\boxed{\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + O(h^2)}$$

**2. What does $O(h^2)$ mean?**

The notation

$$O(h^2)$$

means that the remaining terms are of order $h^2$ or smaller.

For example,

$$\frac{h^2}{6} f'''(x) + \frac{h^3}{24} f^{(4)}(x) + \cdots$$

starts with an $h^2$ term.

So we write all of those higher-order terms compactly as

$$O(h^2).$$

Thus:

$$\boxed{\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2} f''(x) + O(h^2)}$$

**3. Now, what is the error?**

The exact derivative is

$$f'(x)$$

but our numerical approximation is

$$\frac{f(x+h)-f(x)}{h}.$$

So, define the **truncation error** as

$$E = \frac{f(x+h)-f(x)}{h} - f'(x).$$

Using the Taylor expression:

$$E = \frac{h}{2} f''(x) + O(h^2).$$

The leading error term is

$$\frac{h}{2} f''(x).$$

Since this term contains $h^1$, we say

$$\boxed{E = O(h)}$$

Therefore, the **first-order forward difference has first-order truncation error**.

**Three-Point Forward Formula**

The formula

$$f'(x) \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}$$

has truncation error

$$\boxed{O(h^2)}$$

Therefore, it is generally more accurate.

**Second Derivative Formula**

For

$$f''(x) \approx \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2},$$

the truncation error is

$$\boxed{O(h)}.$$